

# TOBB HPC GPU Cluster User Guide

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Acknowledgments

## 1. Overview

Welcome to the TOBB University HPC GPU Cluster! This guide helps you access the cluster, run jobs, and use GPU resources efficiently. All compute nodes run Rocky Linux 8 and feature NVIDIA H100 GPUs.

## 2. System Specs

Cluster machine:

tobb01: 10.1.16.161

You can only connect when you are on the university network.

Key hardware specs (per machine):

CPU: 2 × Intel® Xeon® Gold 6426Y (32 cores total)

RAM: 503 GB

GPU: 2 × NVIDIA H100 PCIe (81 GB VRAM each)

OS: Rocky Linux 8

CUDA Version: 12.4

Slurm Version: 23.11.42

## 3. Accessing the Cluster (SSH)

Connect from within the university network using your hpc credentials:

```
ssh yourusername@10.1.16.161 # tobb01
```

## 4. User Storage and Quota

Default disk quota: 30 GB per user

If you need more space, contact the admin (see Contact).

## 5. Conda Environment Usage

A pre-configured Conda environment (env1) is provided with essential packages for deep learning, computer vision, and NLP.

Activate it before running any job (interactive or via Slurm):

```
module load miniconda
eval "$(conda shell.bash hook)"
conda activate env1
```

Installed packages include (not limited to):

pytorch, torchvision, torchaudio (CUDA 12.4)  
transformers, huggingface-hub  
scikit-learn, pandas, matplotlib, seaborn  
opencv-python  
Full CUDA 12.4 stack

## 6. Slurm: Submitting and Managing Jobs

We use Slurm as the workload manager.

Main partitions:

debug: short, interactive tests (15 minutes max)  
compute: standard jobs, soft time limit (12 hours max., if the job queue is not empty your job will be killed without notice)

How to submit a job:

i. Write a Slurm batch script (see template below).

ii. Submit with:

```
sbatch yourscrip.sh
```

iii. Check your jobs:

```
squeue -u yourusername
```

## 6.1. Example Slurm Script

Create a file `run_job.sh` with the following template:

```
#!/bin/bash
#SBATCH --partition=compute # Partition name: debug or compute #SBATCH
--job-name=myjob # Name for your job
#SBATCH --output=logs/%x_%j.out # Standard output (%x=job name, %j=job ID)
#SBATCH --error=logs/%x_%j.err # Standard error file
#SBATCH --gres=gpu:1 # Number of GPUs (max 1 per node)
#SBATCH --cpus-per-task=4 # Number of CPU cores
#SBATCH --mem=64G # RAM requested
#SBATCH --time=12:00:00 # Time limit hrs:min:sec (debug: max 00:15:00)

# Load modules and activate conda
module load miniconda
eval "$(conda shell.bash hook)"
conda activate env1

# Your Python script or command goes here
python myscript.py
```

Tips:

We are expecting normal usage with single GPU per job. If you need two GPUs because of model size, please contact administrator. But you probably do not.

Set appropriate CPU and RAM according to your job's needs.

Create a `logs/` directory before running your jobs, or change the output paths.

## 7. Troubleshooting & Support

Job not starting: Check partition availability; use `squeue` or `sinfo` to see node/job status.

Out of memory errors: Lower batch size or request more RAM with `--mem=`.

Disk quota exceeded: Clean up old files or request more quota.

GPU not utilized: Ensure CUDA and PyTorch versions are compatible; check your output/error logs.

## 8. Frequently Asked Questions (FAQ)

Q: Which partition should I use?

A: Use debug for short tests (max 15 min), compute for longer/production jobs.

Q: Which conda environment should I use?

A: Using env1 for deep learning, NLP, and CV is recommended as they are already tested. (PyTorch, HuggingFace, OpenCV included).

Q: I need a package that's not installed. What should I do?

A: Not recommended but you can create your own environment in home directory and use that within slurm.

Q: How do I check my disk usage?

A: Run:

```
du -sh ~
```

Q: How can I transfer files to/from the cluster?

A: Use scp or rsync from a machine connected to the university network:

```
scp yourusername@10.1.16.161:/path/to/remote/file /local/dir  
scp /local/dir yourusername@10.1.16.161:/path/to/remote/dir
```

Q: How do I monitor GPU usage?

A: Run:

```
nvidia-smi
```

## 9. Contact

For any issues, contact the cluster admin:

Email: [hpc.tobbetu@gmail.com](mailto:hpc.tobbetu@gmail.com)

Contact: Bahaeddin Eravci

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